

TOXICITY TEST WITH THE MARINE CRUSTACEAN ACARTIA TONSA

Group 8

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ABSTRACT

The aim of this experiment was to determine the acute toxicity of 4-nitrophenol toward the marine crustacean *Acartia tonsa* by performing a short-term lethality test over 24 hours. A concentration series of 0–120 mg/L was tested, with four replicates per concentration and approximately 4–8 organisms per replicate. Mortality was used as the endpoint, and concentration–response relationships were analysed to estimate LC-values.

The results showed a clear concentration-dependent increase in mortality. Low mortality was observed in the control (16.7%) and at 7.5 mg/L (28.6%), while a sharp increase occurred at intermediate concentrations, reaching 96.4% mortality at 30 mg/L and complete mortality at 120 mg/L. The fitted concentration–response curve displayed a typical sigmoidal shape, with a steep transition between 15 and 30 mg/L and a plateau at higher concentrations. The estimated LC10, LC20, and LC50 values were 5.30 mg/L, 7.25 mg/L, and 13.22 mg/L, with 95% confidence intervals of [0.5363, 10.07], [2.314, 12.20], and [8.052, 18.39], respectively. These relatively narrow confidence intervals are indicating good model precision and good agreement between model prediction and measured data.

Although the test formally did not meet the validity criterion due to elevated control mortality (>10%), the clear dose–response relationship and consistent trend across concentrations suggest that the results are still informative. Overall, the study demonstrates that 4-nitrophenol exhibits significant acute toxicity toward *A. tonsa*, with effects occurring at relatively low concentrations. The retrieved values are primarily used for a rapid screening tool to identify acute toxicity, that will later support further investigations and environmental hazard assessment of 4-nitrophenol contamination.

INTRODUCTION

Acartia tonsa is a calanoid copepod commonly recommended by the International Organization for Standardization (ISO) for the evaluation of acute lethal toxicity of contaminants. This test organism meets all the criteria needed for it to be considered a model species: ecological relevance, sensitivity, wide distribution, and easy handling in the laboratory thanks to continuous egg production.

It serves as a representative primary consumer and food source (for secondary consumers such as zooplanktons and planktivorous fish) in ecotoxicological studies and environmental risk assessments in marine and estuarine systems. [1]

The compound 4-nitrophenol is characterized by moderate solubility (16 g/L at 25 °C), indicating high mobility and tendency to remain in the aqueous phase. In contrast, it has low volatility as reflected from a small Henry's constant ($H < 10^{-5}$ atm·m³/mol) and low vapor pressure (< 0.01 mmHg). As a result, the compound is not expected to be partitioned into the gas phase. This makes 4-nitrophenol a safe compound to work with in the lab since inhalation is not expected to be a significant exposure pathway. In addition, volatilization losses are unlikely, as the compound is expected to remain within the experimental vials. Furthermore, the compound has a low bioaccumulation potential (log (Kow) < 4); therefore, additional losses due to accumulation in *A. tonsa* can be considered negligible [2].

The goal of this experiment was to conduct an acute toxicity test on *A. tonsa* to quantitatively determine the lethality effect of 4-nitrophenol (4- NP) on the test organism, also known as mortality rate. The objective of this study was to determine the LC10 and LC50 values, including their associated confidence intervals, by comparing exposed organisms with an unexposed control population. The calculated mortality rates serve as a direct metric for quantifying the lethal effects of the mixture [3].

METHOD

The acute toxicity test with the marine crustacean *A. tonsa* was conducted for five different concentrations of 4-nitrophenol. The concentrations were deduced from an initial EC50 value of 30 mg/L (average of literature findings [4][5]) and accordingly reduced or increased by a certain factor (¼, ½, 1, 2, 4) to have a meaningful range of test values.

- Concentration of the stock solution: 2400 mg/L
- Concentration range used: (0, 7.5, 15, 30, 60, 120) mg/L

The principle of the test was to evaluate the acute toxic effects of the compound based on the mortality rate of the marine copepod *A. tonsa* when exposed to different concentrations.

To initiate the test, five test solutions were prepared in 100 mL volumetric flasks. The 100 mL from each flask was then distributed equally into 4 glass beakers (50 mL capacity), resulting in 4 replicates per concentration (in total 20 glass beakers). To have a basis for comparison, the test also included 4 control samples containing only the growth medium.

Approximately 4 to 8 animals (2-3 weeks old) were carefully transferred into each test beaker. After every transfer, the precise initial count of *A.tonsa* per beaker was recorded (see raw data Table in the Appendix) in order to have the starting count of animals at t= 0h (ranging from 18 to 29 total animals per concentration group). Each beaker was then covered with a thin transparent film to protect the sample and maintain stable conditions, and later they were stored in darkness at room temperature.

After 24 hours, mortality was evaluated by counting the number of dead animals. Animals were considered dead if they exhibited no mobility; in uncertain cases, a metal stick was used to gently stir the surrounding water to verify the lack of movement [3].

Data treatment

The mortality rate (m) in each replicate beaker was calculated using the following equation:

$$m = \frac{N_{dead}}{N_{total}} * 100$$

Where:

- N_{dead} is the number of dead animals after the 24-hour exposure period.
- N_{total} is the exact initial number of animals transferred to the specific beaker.

The concentration-response dataset was analyzed using R software following DTU guidance to generate concentration-response curves and estimate the LC10 and LC50 values along with their 95% confidence intervals.

RESULTS

Starting from the raw data gathered in the *Raw data table* in the Appendix, the following concentration-response data for *A. tonsa* mortality after 24h was derived:

Table 1. Mortality of *A. tonsa* after 24 h exposure to 4-nitrophenol.

Concentration of 4-NP (mg/L)	Total number of animals at start	Total number of dead animals after 24h	Mortality rate, m
0	18	3	16.7 %
7.5	21	6	28.6%
15	25	12	48.0%
30	28	27	96.4%
60	29	28	96.6%
120	22	22	100%

The results show a clear concentration-dependent effect within the tested concentration range.

The total number of dead animals after 24h was 3 in the control and as concentration was increased the number increased, from 6 at 7.5 mg/L to complete mortality of the population at 120 mg/L. Respectively, the mortality % increased from 28.6% to 100%, demonstrating a relevant chronic effect of 4-nitrophenol starting already from the mid-range concentration, specifically from 15 mg/L (see the Discussion section for further considerations).

The concentration-response curve for the mortality after 24h can be observed in *Figure 1* below.

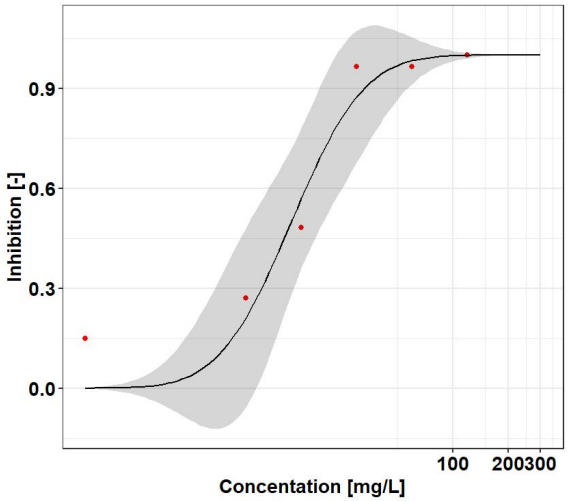


Figure 1: Concentration-inhibition curves after 24h.

The concentration–response curve demonstrates a sigmoidal increase in mortality with rising test concentration in *A.tonsa*. At the lowest concentrations, the mortality rate of the exposed population was minimal, indicating little effect of the 4-nitrophenol. Increasing concentrations resulted in rapid increase in inhibition reflecting a strong dose - dependent response.

The steepest part of the curve occurs at the intermediate concentrations (15 mg/l and 30 mg/l) where small changes of the concentration result in large changes in the lethality, going from 28.6% to 96.4%. At the highest tested concentrations (30 mg/l and 60 mg/l) the curve reaches a plateau, ultimately reaching 100% mortality in the population exposed to 120 mg/l of 4-nitrophenol.

The LC10, LC20, and LC50 values estimated are presented in Table 2, together with their standard deviations (SD) and 95% confidence intervals (CI).

Table 2: EC10, EC20 & EC50 values after 24 h with respectively Confidential interval (CI) and Standard deviation (SD).

	Value (mg/L)	SD	Lower (95% CI)	Upper (95% CI)
LC10	5.301	1.716	0.5363	10.07
LC20	7.255	1.779	2.314	12.20
LC50	13.22	1.862	8.052	18.39

The retrieved results are characterized by narrow confidential intervals and small standard errors in both indicators of high-level precision in the model fitting. This is also further supported by the closeness of the red points (experimentally measured) to the fitted sigmoidal curve indicating a good agreement between the measured data and the model prediction. Together, the statistical results (see Table 2) and the close alignment are good indicators of robustness of the concentration–response relationship for the *A.tonsa* test.

DISCUSSION

Test Validity and Physical-Chemical Stability

The validity of the test is assessed based on the control mortality, which must not exceed 10% [3]. In this study, the control mortality reached 16.7%, meaning that the validity criterion was not fulfilled. This deviation reduces the reliability of the results, as elevated control mortality may indicate stress unrelated to the test

compound. However, more than one partial response was retrieved, and a clear concentration-response relationship could be observed so after consulting with our supervising professor, the test was considered valid to develop this report.

The physicochemical properties of 4-nitrophenol however supports the concentration–response results. The compound has high water solubility and low volatility, meaning it remains dissolved in the aqueous phase and does not evaporate significantly. As a result, the test organisms were likely exposed to stable and homogeneous concentrations throughout the experiment. This also ensures that the deviation between the actual and nominal concentrations can be considered negligible and increases the reliability of the LC values obtained from the study. Furthermore, the final readings may nevertheless be affected by minor factors such as bioavailability and toxicity since 4-NP is characterized by a pKa of 7.15 suggesting that both protonated and deprotonated forms may have been present. Lastly, photodegradation must be considered, as the presence of an aromatic ring enables light absorption and renders the compound photolabile [6]. However, the toxicity test has been conducted under low light conditions, with the beakers stored in the dark. Thus, experimental setup limits the influence of photodegradation on the endpoint values and avoids concentration decreases or by-product formation.

Concentration-Response Analysis and Limitations

Despite the validity test being slightly surpassed, the experiment produced a clear and consistent concentration–response relationship, with mortality increasing systematically with concentration. The sigmoidal shape of the curve and the narrow confidence intervals suggest that the model fitting was robust. Therefore, although the test is formally invalid, the data can still be considered indicative of toxicity trends.

The estimated LC10, LC20, and LC50 values were 5.30, 7.25, and 13.22 mg/L, respectively, which indicate that 4-nitrophenol is acutely toxic to *A. tonsa* under the tested conditions. The steepest part of the curve occurred between 15 and 30 mg/L, indicating that this region was the most sensitive to concentration change, therefore, an informative concentration range for estimating lethal effects. The observed plateau at high concentrations (≥30 mg/L), where mortality approached 100%, reflects saturation of the biological response. At these concentrations, all organisms are affected and further increases in concentration do not result in higher mortality. This plateau is important for reliable LC estimation, as it anchors the upper part of the dose–response curve. In this experiment, the plateau was well captured, which contributed to the relatively precise LC50 estimate.

Several limitations should be acknowledged. First, the elevated control mortality weakens confidence in the absolute LC values obtained. Second, although a full concentration–response curve was achieved, the steep transition between 15 and 30 mg/L suggests that additional intermediate concentrations could improve the precision of LC estimates. A further limitation of this study is the variability in the number of organisms per replicate, which introduces uncertainty into the mortality calculations. The manual transfer and subsequent counting of these small, active copepods could have increased the risk of handling-related stress, visual misclassification of live and dead individuals, and operator-dependent errors, particularly considering the short exposure period. The prolonged transfer process during which organisms were exposed to light, likely further compromised their physiological state before the test began. Consequently, some observed mortality may reflect experimental stress in addition to the toxic effect of 4-nitrophenol.

Comparison with Literature LC-values for 4-Nitrophenol

The estimated LC50 value of 13.22 mg/L indicates acute toxicity to *A. tonsa*. When compared to literature values for other aquatic organisms, such as *Daphnia magna* (LC50 ≈ 24 mg/L, 24 h) [7], the results suggest that *A. tonsa* may be more sensitive to 4-nitrophenol because a lower LC50 corresponds to a stronger toxic effect. This comparison is useful as a general indication of sensitivity, but it should not be treated as a strict

equivalence because the organisms belong to different taxa and the test conditions may differ substantially therefore significantly affecting toxicity outcomes.

Additionally, the DTU validation report for *A. tonsa* confirms that the species is a suitable acute toxicity test organism and reports LC50 values for reference toxicants in standardized testing. That supports the ecological and methodological relevance of using *A. tonsa* in acute toxicity assessment, even if directly comparable 4-nitrophenol data for the same species or species from the same taxa were not found [8].

CONCLUSION

The 24-hour acute toxicity test quantified the toxicity of 4-nitrophenol toward *A. tonsa*. Although the validity criterion was not fulfilled due to elevated control mortality (16.7% > 10%), a clear and consistent concentration–response relationship was observed, indicating that the data remain informative.

Mortality increased systematically with concentration, following a sigmoidal pattern and reaching a plateau at higher concentrations (≥30 mg/L), where complete mortality was observed.

The model fitting enabled the estimation of key ecotoxicological endpoints, with LC10, LC20, and LC50 values of 5.30, 7.25, and 13.22 mg/L, respectively. The relatively narrow confidence intervals suggest good precision despite experimental limitations.

While the concentration range was suitable for estimating LC50, the steep transition between 15 and 30 mg/L indicates that additional intermediate concentrations could improve accuracy. Moreover, elevated control mortality and variability in handling and counting of organisms introduce uncertainty. Overall, the results demonstrate that 4-nitrophenol is acutely toxic to *A. tonsa* and provides useful data for environmental risk assessment.

REFERENCES

[1] G. Gorbi *et al.*, “Standardized methods for acute and semichronic toxicity tests with the copepod *Acartia tonsa*,” Environmental Toxicology and Chemistry, 2012.

[2] National Center for Biotechnology Information (NCBI), “4-Nitrophenol,” *PubChem*, 2026. [Online]. Available: https://pubchem.ncbi.nlm.nih.gov/compound/4-Nitrophenol

[3] K. O. Kusk, “Acute Toxicity Test with the Marine Crustacean *Acartia tonsa*”, lab protocol, updated by S. Kruse and S. N. Sørensen, DTU Environment, Lyngby, Denmark, 2019.

[4] H. Liu, J. Wang, Y. Zhang, and X. Zhang, “Detection of 4-nitrophenol, a model toxic compound, using multi-walled carbon nanotube-based sensors,” *Frontiers in Environmental Science*, vol. 8, Art. no. 5, 2020. [Online]. Available: https://www.frontiersin.org/articles/10.3389/fenvs.2020.00005/full. [Accessed: 2-Mar-2026].

[5] T. Karpouzas, E. Capri, and E. Papadopoulou-Mourkidou, “Ecotoxicological assessment of the herbicide 4-nitrophenol in aquatic systems,” *Chemosphere*, vol. 62, no. 9, pp. 1539–1550, 2006. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0045653505011860. [Accessed: 2-Mar-2026]

[6] A. Baun, N. Nyholm, and K. O. Kusk, Environmental Risk Assessment of Chemicals, Note 1, Technical University of Denmark DTU, 2022, sec. 2.6.

[7] “4-Nitrophenol Safety Data Sheet,” Echemi, [Online]. Available: https://www.echemi.com/sds/4-nitrophenol-pid_Seven1024.html [Accessed: Apr. 26, 2026].

[8] K. O. Kusk et al., "Validation of full life-cycle test with the copepod *Acartia tonsa*," DTU Environment, Lyngby, Denmark, Rep. MR2005-146, 2005. [Online]. Available: https://backend.orbit.dtu.dk/ws/files/6580934/MR2005-146.pdf

APPENDIX

A1. Raw data Table

Concentration of 4-NP [mg/L]	N° live animals at start	Total n° animals	N° alive after 24	N° dead after 24h	Total n° dead	Mortality after 24h
0	4	18	4	0	3	0,167
	5		3	2		
	4		4	0		
	5		4	1		
7.5	5	21	3	2	6	0,286
	7		5	2		
	4		4	0		
	5		3	2		
15	6	25	4	2	12	0,480
	6		2	4		
	7		4	3		
	6		3	3		
30	7	28	1	6	27	0,964
	6		0	6		
	7		0	7		
	8		0	8		
60	7	29	0	7	28	0,966
	7		0	7		
	8		0	8		
	7		1	6		
120	5	22	0	5	22	1.00
	6		0	6		
	5		0	5		
	6		0	6		